

AI Labs : AI Solutions - Ank Kumar

EAAP Enterprise AI Automation Platform

Amazon Bedrock Knowledge Base : BIAW4U4TIL · ap-southeast-2  
Answer from documents : Claude Haiku 4.5 · au.anthropic.claude-haiku-4-5-20251001-v1:0  
Design with the framework : Claude Sonnet 4.5 · au.anthropic.claude-sonnet-4-5-20250929-v1:0

QUERY

Answer from documents

Design with the framework

Applies the framework to a new problem or domain. Framework elements are cited; Claude's own proposals are tagged.

DOMAIN

Transportation

USE-CASE

Fleet Maintainance - Bus

Run query

New conversation

first text 4027 ms · total 56304 ms

Start from a known question

What are the core stages of the Enterprise AI Automation Framework?

How does AAIT implement AI-augmented incident triage?

What governance controls sit between an AI decision and an automated action?

**DESIGN** Propose an AI Solution architecture using the Enterprise AI Automation Framework.

RESPONSE

1

 taken from your documents · 

proposed

 Claude's own design reasoning, not in the documents

AI Solution Architecture: Fleet Maintenance – Bus Transportation

Solution Name: BFMP – Bus Fleet Maintenance Platform

Stage 1: Signal Ingestion 

1

5

**Purpose:** Capture real-time and batch signals from bus fleet operations 1 5

- **Telematics streams** proposed: Engine diagnostics (RPM, temperature, oil pressure), brake wear sensors, transmission health, battery voltage from onboard vehicle systems
- **Driver reports** proposed: Digital defect logs submitted via mobile app at shift end
- **Maintenance management system** proposed: Historical work orders, parts inventory, scheduled PM dates, vehicle age and mileage
- **GPS/route data** proposed: Actual vs. planned route adherence, idle time, harsh braking events
- **Environmental sensors** proposed: Ambient temperature, humidity (impacts HVAC and electrical systems)
- **Passenger door cycle counters** proposed: Actuations per trip (wear indicator)

All signals timestamped and normalized into a common schema before Stage 2 1 5.

## Stage 2: Knowledge Retrieval 1 5

**Purpose:** Retrieve domain-specific context to enrich reasoning 1 5

- **OEM service bulletins and technical manuals** proposed: Manufacturer-specific maintenance procedures, torque specs, diagnostic trouble code (DTC) definitions
- **Fleet maintenance history** proposed: Vector embeddings of past work orders, failure modes, parts replaced, labor hours by vehicle ID and component
- **Regulatory compliance knowledge** proposed: Inspection schedules, brake certification intervals, emissions testing requirements
- **Parts supplier lead times and availability** proposed: Current inventory levels, supplier delivery windows
- **Seasonal patterns** proposed: Historical failure rates by month (e.g., HVAC failures spike in summer, battery failures in winter)

Retrieved documents and records passed to Stage 3 as context 1 5.

## Stage 3: Prompt Construction 1 4 5

**Purpose:** Assemble structured prompts combining signals and knowledge 1 4 5

- **Prompt template** proposed: "Vehicle [ID], [make/model], [mileage] mi, [age] years. Current signals: [DTC codes], [sensor readings]. Driver reported: [defect description]. Maintenance history: [last 5 work orders]. OEM bulletin: [relevant TSB]. Assess: (1) failure risk (critical/high/medium/low), (2) root cause hypothesis, (3) recommended action (immediate repair/schedule PM/monitor), (4) estimated downtime, (5) parts required."
- **Multi-vehicle batch prompts** proposed: For fleet-wide trend analysis, aggregate signals from all buses on same route or with same component age
- **Contextual flags** proposed: Tag vehicles approaching regulatory inspection deadlines, under warranty, or flagged by drivers in past 7 days

Prompts formatted for deterministic parsing of LLM output in Stage 5 [1](#) [4](#) [5](#).

#### Stage 4: LLM / AI Reasoning [1](#) [4](#) [5](#)

**Purpose:** Generate maintenance recommendations and risk assessments [1](#) [4](#) [5](#)

- **Primary reasoning task** [proposed](#): Correlate multi-modal signals (telematics + driver reports + history) to diagnose likely failure modes and prioritize interventions
- **Secondary reasoning task** [proposed](#): Predict optimal maintenance windows based on route schedules, parts availability, and technician capacity
- **Model selection** [proposed](#): Fine-tuned LLM on transportation maintenance corpus (work orders, service manuals, failure reports) to improve domain accuracy
- **Latency consideration:** LLM reasoning sits **outside** the real-time vehicle monitoring path [proposed](#). Real-time DTC alerts (e.g., engine overheat) trigger deterministic safety protocols (immediate pull-over notification) without waiting for LLM inference [proposed](#).

Output: unstructured text containing risk level, diagnosis, recommended actions, and justification [1](#) [4](#) [5](#).

#### Stage 5: Structured Decision [1](#) [4](#) [5](#)

**Purpose:** Parse LLM output into executable maintenance work orders [1](#) [4](#) [5](#)

- **Extraction logic** [proposed](#): Regex + JSON schema validation to extract: {vehicle\_id, risk\_level, failure\_mode, recommended\_action, parts\_list[], estimated\_labor\_hours, confidence\_score, reasoning\_summary}
- **Prioritization scoring** [proposed](#): Combine risk level + vehicle utilization + regulatory deadline proximity into a numeric priority score (0-100)
- **Fallback handling** [proposed](#): If LLM output is unparseable or missing required fields, flag for human review and do not auto-generate work order
- **Integration format** [proposed](#): Structured decision formatted as API payload for maintenance management system (e.g., work order creation endpoint)

Output: machine-readable decision object ready for governance evaluation [1](#) [4](#) [5](#).

#### Stage 6: Confidence / Governance [1](#) [4](#) [5](#)

**Purpose:** Apply governance controls to determine execution path [1](#) [4](#) [5](#)

**Confidence thresholds** [1](#) [5](#):

- **High confidence ( $\geq 0.85$ )** [proposed](#): Routine maintenance (oil change, tire rotation) with clear diagnostic codes and matching historical patterns → Automated Path
- **Medium confidence (0.60–0.84)** [proposed](#): Component replacement recommendations (alternator, brake pads) with ambiguous signals → Human Approval
- **Low confidence ( $< 0.60$ )** [proposed](#): Complex diagnostics (intermittent electrical faults) or safety-critical systems (brakes, steering) → Human Approval

**Governance rules** [1](#) [5](#):

- **Safety-critical systems** proposed: Any recommendation affecting brakes, steering, or structural integrity requires human approval regardless of confidence score
- **Cost threshold** proposed: Repairs exceeding \$5,000 or requiring vehicle out-of-service >48 hours require fleet manager approval
- **Regulatory compliance** proposed: Inspection-related actions auto-approved if within 30 days of deadline and confidence  $\geq 0.80$
- **Audit trail** 1 5: All decisions (automated or human-approved) logged with reasoning summary, confidence score, approver ID, and timestamp

**Branch logic** 1 4 5: High-confidence + non-safety-critical + below cost threshold → Automated Path; all others → Human Approval path.

### Stage 7: Execution & Feedback 1 4 5

#### Automated Path 1 4 5:

- **Work order creation** proposed: API call to maintenance management system to generate work order with pre-filled vehicle ID, parts list, labor estimate, and priority
- **Parts reservation** proposed: Automatic inventory hold for required components
- **Technician assignment** proposed: Route to next available technician with required skill certification
- **Schedule optimization** proposed: Slot maintenance during off-peak hours or when vehicle returns from route

#### Human Approval Path 1 4 5:

- **Review dashboard** proposed: Fleet manager sees vehicle ID, risk summary, LLM reasoning, recommended action, and confidence score
- **Approval actions** proposed: Approve (triggers automated execution), Modify (adjust parts/labor), Reject (return to monitoring), Escalate (send to senior technician for physical inspection)
- **Approval SLA** proposed: Medium-risk items reviewed within 4 hours; low-confidence items within 24 hours

#### Feedback capture 1 5:

- **Outcome logging** proposed: Actual parts used, labor hours, root cause confirmed/refuted, vehicle return-to-service time
- **Technician annotations** proposed: Free-text notes on whether AI recommendation was accurate, what was missed, what was over-specified

### Stage 8: Offline Improvement 1 5

**Purpose:** Continuously refine models and knowledge base 1 5

- **Model retraining** proposed: Monthly fine-tuning on new work orders where technician confirmed or corrected AI diagnosis

- **False positive analysis** proposed : Review cases where AI recommended repair but technician found no issue; adjust confidence calibration
- **Knowledge base updates** proposed : Ingest new OEM bulletins, add embeddings for novel failure modes observed in past quarter
- **Threshold tuning** proposed : Adjust Stage 6 confidence thresholds based on precision/recall metrics (target: <5% unnecessary work orders, <2% missed critical failures)
- **Simulation** 1 5 : Replay historical signal data through updated model to validate improvement before production deployment

No live A/B testing on safety-critical decisions proposed ; simulation-only validation for brake/steering recommendations.

### Assumptions

- Buses are equipped with telematics units providing real-time sensor data and DTC codes proposed
- Maintenance management system has API for programmatic work order creation proposed
- Fleet operates on fixed routes with predictable schedules, enabling maintenance window optimization proposed
- Technicians have mobile devices for digital defect reporting and work order updates proposed
- Historical maintenance records are digitized and structured (not paper-based) proposed
- LLM inference latency is 2–5 seconds, acceptable for non-emergency maintenance planning proposed
- Parts inventory system is integrated and provides real-time availability data proposed

### What to Validate with the Client

- **Regulatory jurisdiction and inspection requirements** proposed : Specific intervals, certification bodies, and documentation standards vary by region
- **Safety-critical system definitions** proposed : Client's internal policy on which components require mandatory human approval
- **Cost and downtime thresholds** proposed : Dollar limits and out-of-service duration that trigger escalation to fleet manager or finance
- **Technician skill certification matrix** proposed : Which repairs require ASE-certified mechanics vs. general technicians
- **Union or labor agreements** proposed : Constraints on automated work assignment or scheduling
- **Existing maintenance management system** proposed : API capabilities, data schema, integration effort

- **Telematics vendor and data format** proposed: Real-time vs. batch, protocol (CAN bus, J1939), data completeness
- **Acceptable false positive/negative rates** proposed: Client's risk tolerance for unnecessary repairs vs. missed failures
- **Human approval SLA expectations** proposed: How quickly fleet managers can review and decide on flagged recommendations
- **Data retention and audit requirements** proposed: How long to store reasoning traces, who has access, compliance with transportation authority audits

SOURCES

BMRS\_Solution\_Architecture.docx [https://ank-aws-learning-144160833556.s3.ap-southeast-2.amazonaws.com/ey-fde/BMRS\\_Solution\\_Architecture.docx](https://ank-aws-learning-144160833556.s3.ap-southeast-2.amazonaws.com/ey-fde/BMRS_Solution_Architecture.docx)

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TRACE

|                                 |                                       |                                                                                                   |                                                       |                                      |                               |
|---------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------|-------------------------------|
| USER QUERY<br>157 chars         | KNOWLEDGE ROUTING<br>design · top_k 5 | BEDROCK KNOWLEDGE BASE<br>Retrieve x2<br>1790 ms                                                  | RETRIEVED CHUNKS<br>8 passages                        | CLAUDE<br>ConverseStream<br>54512 ms | GROUNDED RESPONSE<br>end_turn |
| MODE<br>design                  |                                       | TOTAL LATENCY<br>56304 ms                                                                         | TIME TO FIRST TOKEN<br>2237 ms                        |                                      |                               |
| RETRIEVAL<br>1790 ms            |                                       | GENERATION<br>54512 ms                                                                            | RESPONSE STATUS<br>200 OK                             |                                      |                               |
| KNOWLEDGE BASE ID<br>BIAW4U4TIL |                                       | RETRIEVAL OPERATION<br>bedrock-agent-runtime:Retrieve x2 (question + framework reference) top_k=5 | MODEL<br>au.anthropic.claude-sonnet-4-5-20250929-v1:0 |                                      |                               |
| TOKENS IN / OUT<br>2945 / 2443  |                                       |                                                                                                   |                                                       |                                      |                               |

RETRIEVED CHUNKS

[1]

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</relationships> <description> This is a process flow diagram that illustrates the core operational pipeline of an Enterprise AI Automation Framework. The diagram shows six numbered sequential and branching stages that form a reference architecture: **\*\*Linear Sequence (Steps 3-5):\*\*** The first three stages operate in strict sequential order: - Step 3 (PROMPT CONSTRUCTION): The initial stage where input prompts are formulated and prepared - Step 4 (LLM / AI REASONING): The constructed prompts are processed through Large Language Models or AI reasoning engines - Step 5 (STRUCTURED DECISION): The AI output is transformed into structured, actionable decisions **\*\*Decision Point (Step 6):\*\*** Step 6 (CONFIDENCE / GOVERNANCE) serves as a critical control gate where the system's confidence level and governance requirements are evaluated. This stage determines how the decision will be executed. **\*\*Execution Paths:\*\*** At the governance stage, the process splits into two mutually-possible paths: - HUMAN APPROVAL: Represents scenarios where

**[2]** Enterprise\_AI\_Automation\_Framework.docx

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ENTERPRISE AI AUTOMATION FRAMEWORK (EAAF) 1. Purpose The Enterprise AI Automation Framework (EAAF) is a domain-independent architecture for transforming enterprise signals and knowledge into governed AI-assisted decisions and executable business actions. The framework provides the reusable architectural pattern underlying six domain-specific reference implementations: 1. AAIT AI-Augmented Incident Triage 2. SCDO Supply Chain Demand Optimization 3. AAMP - Automotive AI Maintenance Platform 4. FDAB Fraud Detection Architecture Banking 5. TANAOP - Telecom AI Network Assurance and Optimization Platform 6. BMRS Battlefield Munitions Replenishment System The framework separates reusable AI and automation capabilities from the domain-specific business processes, systems, policies, knowledge, and execution mechanisms of each solution. Architectural position EAAF is the reusable architecture. The six solutions are independent peer reference implementations; they are not sequential stages, branches, or

**[3]** Enterprise\_AI\_Automation\_Framework.pdf

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<analysis> <image\_type> Process Flow Diagram / Data Flow Diagram </image\_type> <title> Enterprise AI Automation Framework - Reference Architecture Flow Steps 3-6 </title> <caption> Sequential process flow diagram showing the core stages of an Enterprise AI Automation Framework,

from prompt construction through offline improvement, with a decision point at the governance stage that branches into human approval or authorized execution paths. </caption> <extractions> - Step 3: PROMPT CONSTRUCTION - Step 4: LLM / AI REASONING - Step 5: STRUCTURED DECISION - Step 6: CONFIDENCE / GOVERNANCE - HUMAN APPROVAL (left branch from Step 6) - AUTHORIZED EXECUTION (right branch from Step 6) - EXECUTION & FEEDBACK (convergence point) - OFFLINE IMPROVEMENT (final step) - Downward arrows connecting sequential steps - Diagonal arrows showing branching at Step 6 - Diagonal arrows showing convergence at EXECUTION & FEEDBACK </extractions> <relationships> 1. Linear sequential flow: Steps 2, 4, and 5

[5] Enterprise\_AI\_Automation\_Framework.pdf

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Data Accumulation: Information collected at each stage (signals, knowledge, prompts, reasoning outputs) builds upon and informs subsequent stages. Stage 6 Decision Point: Stage 6 (Confidence/Governance) branches into two parallel paths - "Automated Path" and "Human Approval" - both converging at Stage 7, indicating conditional logic based on confidence levels and governance rules. Feedback Loop: Stage 8 (Offline Improvement) connects back to earlier stages conceptually, suggesting that analysis and simulation results inform future iterations of the model and knowledge base. Input Sources: Stages 1-3 represent input preparation (ingestion, retrieval, construction). Processing Stage: Stage 4 represents the core AI reasoning and analytical processing. Output Stage: Stage 5 generates the structured decision output. Control Stage: Stage 6 applies governance controls and determines execution path. Implementation Stage: Stage 7 executes the decision and captures outcomes. Learning Stage: Stage 8 continuously improves the system based

[6] Enterprise\_AI\_Automation\_Framework.pdf

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The Enterprise AI Automation Framework (EAAF) is positioned at the top as the parent/primary framework 2. All six domain-specific implementations (AAIT, SCDO, AAMP, FDAB, TANAOP, BMRS) are subordinate to and derived from the EAAF 3. Each implementation is connected via a downward arrow, indicating a top-down hierarchical relationship 4. Each acronym corresponds to a specific industry domain: IT Operations, Supply Chain, Automotive, Banking, Telecom, and Defence 5. Each domain implementation includes a parenthetical description of its specific use case or application area 6. The implementations are arranged horizontally in a linear sequence, suggesting they are parallel instantiations of the same framework applied to different domains 7. The framework relationship indicates that all six solutions instantiate the same reusable patterns (orchestration, reasoning, governance, execution, outcome, and continuous-improvement patterns) while customizing the domain context </relationships> <description> This is a hierarchical categorization

[7] Enterprise\_AI\_Automation\_Framework.docx

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Each implementation is connected to the parent framework through downward-pointing arrows, indicating a hierarchical derivation relationship. Each of the six implementations is positioned at an equal level horizontally, suggesting they are parallel, independent instantiations of the same framework adapted to different domains. The framework pattern is applied consistently across all six domains: IT Operations, Supply Chain Management, Automotive, Banking, Telecom, and Defence. Each domain solution addresses a specific use case within its industry vertical. </relationships> <description> This is a hierarchical diagram illustrating the framework-to-solution relationship of the

Enterprise AI Automation Framework (EAAF). The diagram uses a tree-like structure with a single parent node at the top and six child nodes arranged horizontally at the bottom level. At the top of the diagram is the "ENTERPRISE AI AUTOMATION FRAMEWORK (EAAF)" label contained within a rectangular box. A horizontal line extends from this box with six evenly-spaced downward-pointing

[8]

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<analysis> <image\_type> Hierarchical and categorization diagram </image\_type> <title> Enterprise AI Automation Framework (EAAF) - Domain-Specific Reference Implementations </title> <caption> A hierarchical diagram showing the Enterprise AI Automation Framework (EAAF) at the top level, with six domain-specific reference implementations positioned below it, each representing a different industry vertical and use case application. </caption> <extraction> 1. Title: "ENTERPRISE AI AUTOMATION FRAMEWORK (EAAF)" 2. Six domain-specific implementations (acronyms): - AAIT - SCDO - AAMP - FDAB - TANAOP - BMRS 3. Domain categories and associated use cases: - AAIT: IT Operations (Incident Triage) - SCDO: Supply Chain (Demand/Procurement) - AAMP: Automotive (Maintenance) - FDAB: Banking (Fraud) - TANAOP: Telecom (Network Assurance) - BMRS: Defence (Munitions Replenishment) 4. Structural elements: - Horizontal line connecting all implementations - Downward arrows pointing from EAAF to each implementation - Three-level hierarchical structure